

NETWORK TRAFFIC ANALYSIS REPORT

NetSupport Manager RAT — C2 Traffic Investigation

Incident Date: 2026-02-28

Report Date: 2026-06-10

PCAP Source: Malware-Traffic-Analysis.net — 2026-02-28 Exercise

Classification: **CONFIDENTIAL — For Internal Use Only**

1. Executive Summary

On 2026-02-28, a Windows workstation (10.2.28.88 — DESKTOP-TEYQ2NR, user Becka Rolf) was compromised and observed communicating with confirmed malicious infrastructure. Analysis of the captured network traffic reveals the host was infected with NetSupport Manager RAT, a commercially available remote-access tool routinely weaponized by threat actors for post-compromise command-and-control. The infected machine transmitted repeated outbound HTTP POST requests to an attacker-controlled server at 45.131.214.85, associated with the suspicious domain vadusa.xyz and the endpoint /fakeurl.htm — all of which were independently confirmed as malicious by multiple threat intelligence platforms.

The compromise represents an active remote-access threat. The attacker maintained a persistent, low-noise C2 channel using legitimate-looking HTTP traffic, which could facilitate data exfiltration, lateral movement, or further payload delivery. Immediate isolation of the affected host and remediation of the NetSupport RAT infection are recommended alongside a review of related user accounts and network segments.

2. Investigation Overview

2.1 PCAP Source & Timeline

Source: Malware-Traffic-Analysis.net — 2026-02-28 Traffic Analysis Exercise

Capture Date: 2026-02-28

Report Date: 2026-06-10

Analyst: SOC Analyst (Trainee Exercise)

2.2 Tools Used

The following tools were used during this investigation:

- Wireshark — Primary packet capture analysis (Statistics, Conversations, DNS, HTTP filters, TCP stream follow)
- VirusTotal — IOC enrichment for IP addresses, domains, and URLs
- AbuseIPDB — IP reputation and abuse history lookup
- AlienVault OTX — Threat intelligence pulse lookup and passive DNS
- Shodan — Internet-facing infrastructure fingerprinting

2.3 Investigation Methodology

Analysis began with a broad review of endpoint and conversation statistics in Wireshark to identify the host generating the highest volume of traffic. That host (10.2.28.88) was then traced through DNS, HTTP POST, and TCP stream analysis to identify C2 infrastructure. All identified IOCs were enriched across four external threat intelligence platforms, and a malware family was identified based on observed User-Agent strings and traffic patterns. Findings were validated against the published answer key for this exercise.

3. Infected Host Details

The following victim host details were confirmed through packet capture analysis and validated against the exercise answer key:

Infected Host	10.2.28.88
Malware Family	NetSupport Manager/1.3 RAT
Suspicious Infrastructure	45.131.214.85, vadusa.xyz, http://45.131.214.85/fakeurl.htm
Hostname	DESKTOP-TEYQ2NR
MAC Address	00:19:d1:b2:4d:ad
Windows Account	brolf
Full Name	Becka Rolf
OS Platform	Windows (internal subnet 10.2.28.0/24)

⚠ This host should be considered fully compromised. Immediate isolation from the network is recommended pending reimaging and forensic review.

4. Traffic Analysis Findings

4.1 Endpoint & Conversation Analysis

Wireshark's Statistics → Endpoints (IPv4) view was used to identify the host with the highest packet count in the capture. The host 10.2.28.88 stood out as the most active internal address, with the greatest number of packets transmitted and received. This was further confirmed in the Conversations view, where 10.2.28.88 appeared as the source in the highest-volume sessions.

The top external IPs contacted by this host included:

- 45.131.214.85 — Primary C2 server (confirmed malicious)
- 23.218.232.148
- 45.131.214.74
- 150.171.28.11
- 23.192.223.23

The majority of suspicious activity was concentrated in the connection to 45.131.214.85.

4.2 DNS Analysis

Filtering Wireshark for dns revealed multiple DNS queries originating from 10.2.28.88. Among the observed traffic:

Suspicious Query: `vadusa.xyz`

Resolved IP: `45.131.214.85`

The domain `vadusa.xyz` is anomalous for several reasons: it uses the `.xyz` TLD (frequently associated with disposable malicious infrastructure), was registered only three months before the capture date, and resolves to an IP address in the MHost LLC / LeaseWeb hosting ecosystem with no legitimate business presence.

Additional DNS failures were observed for WPAD-related and internal-looking domains:

- `wpad.easyas123.tech` — Repeated NXDOMAIN responses
- `wpad.mshme.net` — Repeated NXDOMAIN responses
- `EASYS123-DC.easyas123.tech` — Likely a domain controller in the lab environment

These WPAD failures, while not confirmed malicious, indicate the host was attempting to discover proxy configuration. Repeated NXDOMAIN failures can also be a sign of poorly configured or adversary-controlled DNS names. They are documented here as anomalous activity requiring further review.

Expected/benign domains also observed in the traffic included: `microsoft.com`, `bing.com`, `windowsupdate.com`, `adobe.com`, `digicert.com`, and `googleapis.com`. These were excluded from the IOC list.

4.3 HTTP POST Analysis

Applying the Wireshark filter `http.request.method == "POST"` revealed multiple HTTP POST requests from 10.2.28.88 to the external IP `45.131.214.85`. Key findings:

Destination IP: `45.131.214.85`

Destination Port: `443`

Suspicious URL: `http://45.131.214.85/fakeurl.htm`

HTTP User-Agent: `NetSupport Manager/1.3`

Example Packet: No. 2638

NetSupport Manager is a legitimate commercial remote-access product. However, its User-Agent string appearing in outbound POST traffic to a known-malicious IP is a strong behavioral indicator that this tool has been deployed as a RAT (Remote Access Trojan) — a technique frequently used by threat actors to abuse trusted software and blend C2 traffic with normal-looking HTTP communication.

The repeated nature of the POST requests (multiple packets over the session duration) is consistent with heartbeat or polling traffic characteristic of a C2 channel, where the implant checks in with the controller at regular intervals.

4.4 TCP Stream Analysis

Following TCP stream 40 (which contained packet 2638) confirmed the complete HTTP conversation. The POST to /fakeurl.htm was transmitted in cleartext HTTP despite port 443 being listed as the destination, which is unusual. This pattern — using a non-standard port or mismatched port/protocol — is a common evasion technique to confuse simple port-based detection rules.

The request content and response were consistent with NetSupport Manager RAT C2 protocol behavior as documented in open-source threat intelligence.

5. Indicators of Compromise (IOC Table)

All indicators identified during this investigation are summarized below. These IOCs should be added to your SIEM, EDR, DNS filtering, and firewall block lists.

Type	Value	Description	Source
IP Address	45.131.214.85	Confirmed C2 server. Destination for repeated HTTP POST requests from infected host. Flagged malicious by VirusTotal (11/91). Hosted Python/BaseHTTP web service on port 80.	Wireshark, VirusTotal, Shodan, OTX
Domain	vadusa.xyz	Suspicious .xyz domain resolving to 45.131.214.85. Flagged malicious by 12/91 VT vendors. Created 2026-02-27 via Dynadot/Dyna-NS. Associated with known malware files.	Wireshark DNS, VirusTotal, OTX
URL	http://45.131.214.85/fakeurl.htm	Repeated HTTP POST endpoint. Flagged malicious by 14/95 VT vendors. Categorized as known infection source / spyware and malware.	Wireshark HTTP POST, VirusTotal
User-Agent	NetSupport Manager/1.3	HTTP User-Agent string observed in POST traffic. Indicates NetSupport RAT C2 communication protocol.	Wireshark HTTP stream
File	usa.confirm.exe	Communicating file associated with IP 45.131.214.85 and vadusa.xyz per VirusTotal relations.	VirusTotal
File Hash	6b80c8998e3ab193824a2dd4a4443447	MD5 hash of sample linked to blankgrabber / glassworm / icedid / luca-stealer / njrat malware families. Associated with C2 infrastructure.	VirusTotal

6. Threat Intelligence Enrichment

6.1 VirusTotal

All three primary IOCs were submitted to VirusTotal and returned positive detections:

- IP 45.131.214.85: Flagged malicious by 11 of 91 vendors. Community score: -12. Linked to vadusa.xyz and known malware samples.
- Domain vadusa.xyz: Flagged malicious by 12 of 91 vendors. Category: Malicious. Created 2026-02-27. Linked to communicating files including usa.confirm.exe and a sample associated with blankgrabber, glassworm, IcedID, Luca Stealer, and NjRAT.
- URL <http://45.131.214.85/fakeurl.htm>: Flagged malicious by 14 of 95 vendors. Community score: -1. Categories: Malicious, Known infection source, Spyware and malware. First submitted 2026-02-28.

6.2 AbuseIPDB

AbuseIPDB returned an Abuse Confidence Score of 0% for 45.131.214.85 with no prior abuse reports. The IP was categorized as a Data Center / Web Hosting host under MHost LLC (ASN200823), located in Frankfurt am Main, Germany. The absence of AbuseIPDB reports likely reflects the IP's recent deployment — VirusTotal first saw the URL on the same day as this capture (2026-02-28), suggesting newly registered infrastructure. The IP remains a confirmed IOC based on multiple other sources.

6.3 AlienVault OTX

OTX confirmed the infrastructure relationship between 45.131.214.85 and vadusa.xyz through passive DNS resolution records. The IP returned HTTP scan data showing a Python 3.12.13 / BaseHTTP web server, consistent with a lightweight C2 server. The domain vadusa.xyz was registered 2026-02-27 through Dynadot with Dyna-NS nameservers, and was resolved to the C2 IP. No threat intelligence pulses were present for either IOC at the time of analysis, consistent with newly deployed infrastructure.

6.4 Shodan

Shodan confirmed 45.131.214.85 had port 80/tcp open as of 2026-05-28, running Python 3.12.13 / BaseHTTP/0.6 and returning HTTP 404 responses. The host was attributed to RapidSeedbox Ltd / LeaseWeb Netherlands B.V. (ASN60781) in Lelystad, Netherlands. No Shodan results were returned for vadusa.xyz directly or the full URL path — consistent with Shodan not indexing path-level URL data.

7. Malware Identification

7.1 Identified Malware Family

Malware Family: **NetSupport Manager RAT**

7.2 Evidence Supporting Identification

The malware identification was based on the convergence of multiple independent indicators:

- HTTP User-Agent string NetSupport Manager/1.3 observed directly in Wireshark packet capture — the clearest single indicator.
- Repeated HTTP POST requests to /fakeurl.htm, a well-documented C2 endpoint associated with NetSupport Manager RAT deployments in threat intelligence reporting.
- VirusTotal relations on vadusa.xyz and 45.131.214.85 linked to a malware sample with the hash 6b80c8998e3ab193824a2dd4a4443447, tagged with multiple malware families including references consistent with RAT tooling.
- Infrastructure characteristics (newly registered domain, Python-based HTTP server, hosting in anonymity-friendly infrastructure) are consistent with threat-actor-deployed NetSupport RAT C2.

7.3 About NetSupport Manager RAT

NetSupport Manager is a legitimate commercial remote-access product sold by NetSupport Ltd. However, threat actors frequently abuse it as a RAT by deploying it on victim machines without authorization — a technique that helps evade detection because the underlying tool is trusted software with a valid digital signature. NetSupport Manager RAT campaigns have been widely documented by threat intelligence researchers and are commonly distributed via malicious email attachments, fake software updates, and drive-by downloads.

Capabilities enabled by NetSupport Manager RAT include remote desktop viewing and control, file transfer, keystroke logging, screen capture, process listing, and arbitrary command execution — representing a complete remote-access toolkit for an attacker.

This malware family identification was validated against the Malware-Traffic-Analysis.net answer key for the 2026-02-28 exercise and confirmed correct.

8. Recommendations

8.1 Immediate Actions (0–24 Hours)

- Isolate 10.2.28.88 (DESKTOP-TEYQ2NR) from the network immediately to stop active C2 communication.
- Preserve a forensic image of the infected host's disk and memory before remediation.
- Reset the credentials of the affected Windows account (brolf / Becka Rolf) across all systems.
- Block 45.131.214.85 and vadusa.xyz at the perimeter firewall, DNS filter, and proxy/NGFW immediately.
- Block the URL <http://45.131.214.85/fakeurl.htm> at the web proxy layer.

8.2 Short-Term Remediation (1–7 Days)

- Reimage the affected workstation — NetSupport Manager RAT installations may persist across simple antivirus removal.

- Hunt across the environment for other hosts showing DNS queries to vadusa.xyz or HTTP POST traffic to 45.131.214.85 — lateral spread should be ruled out.
- Investigate the initial infection vector: review email logs, browser history, and recent downloads on DESKTOP-TEYQ2NR around 2026-02-28.
- Search SIEM/EDR telemetry for the file hash 6b80c8998e3ab193824a2dd4a4443447 and filename usa.confirm.exe across all endpoints.
- Review NetSupport Manager software inventory — if NetSupport Manager is not an authorized business tool, add it to the application blocklist.

8.3 Long-Term Hardening Recommendations

- Enable DNS-over-HTTPS or deploy a DNS security solution (e.g. Cisco Umbrella, Cloudflare Gateway) to block newly registered malicious domains at the resolver level.
- Enforce application control policies to block unauthorized remote-access tools including unapproved versions of NetSupport Manager, AnyDesk, TeamViewer, and similar.
- Implement TLS inspection on outbound traffic where feasible to detect C2 traffic masquerading as HTTPS.
- Subscribe to threat intelligence feeds to receive timely IOC updates and block emerging C2 infrastructure proactively.
- Conduct user security awareness training on phishing and drive-by download threats, as these are the primary distribution vectors for NetSupport Manager RAT.

9. Analyst Notes & Reflection

This report was produced as part of a structured traffic analysis exercise based on a PCAP from Malware-Traffic-Analysis.net. The following items were identified and documented correctly from Wireshark analysis prior to reviewing the answer key:

- Infected host: 10.2.28.88
- C2 IP address: 45.131.214.85
- Malicious domain: vadusa.xyz
- C2 URL: <http://45.131.214.85/fakeurl.htm>
- Malware family: NetSupport Manager RAT

The following victim host details were not documented in the initial analysis and were obtained from the answer key:

- Hostname: DESKTOP-TEYQ2NR
- MAC Address: 00:19:d1:b2:4d:ad
- Windows Account: brolf
- Full Name: Becka Rolf

In a real incident, hostname, MAC address, and user account details would typically be extracted from DHCP server logs, Active Directory, or from the PCAP itself (NBNS/NetBIOS traffic, Kerberos authentication packets, or DHCP lease packets). This should be included as a standard step in future analysis workflows.

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